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Investigation of Main Bearing Fatigue Estimate Sensitivity to Synthetic Turbulence Models Using a Novel Drivetrain Model Implemented in OpenFAST

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ABSTRACT

A coupled medium-fidelity drivetrain model is developed and implemented in OpenFAST for a 10-MW land-based reference turbine. The implementation is verified against a fully coupled multibody wind turbine model, including a detailed drivetrain. The new model can simultaneously and accurately estimate main bearing loads and represent elastic bending of the drivetrain. It has low computational cost and is useful for early design phases, sensitivity analyses and complex systems like wind farms (where computational expense must be expended elsewhere). Here, the model is implemented for a monopile offshore wind turbine and used to investigate the sensitivity of main bearing basic rating life to different synthetic turbulence models. Large-eddy simulations (LES) targeting stable, neutral, and unstable atmospheric conditions at below-, near- and above-rated wind speeds are used as a reference. The turbulence models recommended by the International Electrotechnical Commission, the Mann spectral tensor model, and the Kaimal spectral model with exponential coherence are fitted to the LES data. Additionally, a constrained turbulence generator, PyConTurb (short for *Python Constrained Turbulence*), based on LES data, is applied in the aero-hydro-servo-elastic simulations. Taking PyConTurb as the baseline, the Kaimal model significantly underestimates fatigue of the downwind main bearing, with between 10% and 40% less damage. The Mann model also underestimates the downwind main bearing fatigue by up to 30%. The upwind main bearing damage is driven by mean loads, and differences between models are less significant, although the trends are similar. Reasons for these discrepancies are investigated and attributed to differences in spatial and temporal variations among the turbulence models.

1 | Introduction

Offshore wind turbines are expected to contribute substantially to renewable energy in the years to come, and increasing turbine size is regarded as a means to reduce cost. As rotor diameters

increase, both the aerodynamic loads and the uncertainties resulting from the traditional ways of modeling turbulent wind also increase. The International Electrotechnical Commission (IEC) recommends two synthetic turbulence models for dynamic analyses of wind turbines [1]: the Kaimal spectrum with

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exponential coherence [2] (Kaimal) and the Mann spectral tensor model [3, 4] (Mann). Both are based on neutral atmospheric conditions. However, through the last decade, researchers have shown that their applicability is questionable for several reasons. Offshore conditions change with the season, are seldom neutral, and their characteristics are important for turbine loads [5–9]. The synthetic turbulence models do not account for buoyancy-driven turbulence and give large differences in responses, both for floating [7, 10–13] and bottom-fixed turbines [14, 15]. It is expected that the accurate modeling of turbulent inflow conditions will be more important as turbine size increases.

The aforementioned studies looked into variations in loads on the tower, blades and mooring lines, but there is little focus on the influence of turbulence models on drivetrain components. Main bearings are essential in the load transfer between the rotor and the tower and have recently received increased attention. A field study conducted by Hart et al. [16] revealed that out of 15 GW of installed wind energy, 10% of main bearings on primarily land-based wind turbines under 2 MW in size were replaced after half of a typical 20-year design life. In contrast, Kenworthy et al. [17] only calculated rating lives this low in cases of high operating temperatures and severe lubricant contamination using the Kaimal model with parameters from the IEC standard. However, as will be shown in the coming sections, the Kaimal model does not necessarily provide realistic bearing loads.

Currently, the drivetrain and main bearings are typically considered in a one-way-coupled fashion when evaluating drivetrain structural response, with loads and motions from aero-hydro-servo-elastic (global) analyses provided as input to a detailed drivetrain multibody model. In most global analysis tools, such as OrcaFlex, SIMA, HAWC2 and OpenFAST, the drivetrain can only be included as a torsional spring and damper. It is possible to make a coupled model in multibody software, including other drivetrain degrees of freedom, but the computational effort is significant, which makes it unfeasible to run the numerous load cases required by industry standards. Another justification for decoupling has been the separation of ranges of natural frequencies associated with the drivetrain compared to the aeroelastic domain. As offshore wind turbines increase in size, the separation between ranges of natural frequencies may decrease, thus increasing the need for coupled analysis. Torsvik et al. [18, 19] highlighted the importance of considering large bearings as flexible components and argued that fully coupled drivetrain analyses may be necessary to capture dynamics properly. Using a multibody tool, Wang et al. [20] discovered that a fully coupled model, including flexible drivetrain components and bedplate, led to nontorsional drivetrain modes at lower natural frequencies than the first torsional mode. Moreover, Wang et al. [21] found that including a flexible bedplate in global analyses of a 10-MW floating turbine reduced the damage equivalent fatigue loads of the upwind main bearing by up to 37% and conversely increased the damage equivalent fatigue loads of the downwind main bearing by up to 54%. The power-to-weight ratios of drivetrains are expected to increase with turbine size, further increasing the need to improve drivetrain structural analysis [22].

Wind turbines experience significant variations in aerodynamic loading due to wind fluctuations. Based on experimental data from a 6-MW floating wind farm, Torsvik et al. [23] discovered that main bearing axial and radial loads are driven by

differential out-of-plane bending moments between the blades at three times the blade passing frequency (3P). Similarly, Lavelly [24] found nontorque shaft bending moments to be potential drivers for premature bearing fatigue failure and that these bending moments were caused by spatial variations in the wind velocity across the rotor.

This work presents a novel drivetrain model implemented in the aero-hydro-servo-elastic software OpenFAST. The purpose of the model is to capture bending degrees of freedom of the drivetrain and obtain reasonably accurate main bearing loads with low computational cost. Drivetrain load formulations presented by Guo et al. [25, 26] and Bergua et al. [27] and main bearing models made by Hart et al. [28, 29] and Stirling et al. [30] were used as guidelines in this work. The implemented model is simple but more detailed than the typical torsional spring representation and allows for extraction of bearing loads as well as modeling of flexible components through beams and massless springs. The 10-MW land-based reference wind turbine (RWT) from the Technical University of Denmark (DTU) [31] was considered, with the original medium-speed drivetrain and moment supporting main bearings developed by Wang et al. [32]. The model was verified against a fully coupled multibody model, including a detailed drivetrain with flexible components, and flexible tower and blades. This verification suggests that the model can accurately reproduce global and local response, particularly the main bearing loads. The new model and verification are presented in Section 2.

In Section 3, the coupled drivetrain model is applied to the bottom-fixed offshore version of the DTU 10-MW RWT with simply supported main bearings. The low computational cost allows investigation of several realizations and environmental conditions and the resultant main bearing radial and axial loads. Large-eddy simulations (LES) were used to generate canonical cases of stable, neutral and unstable atmospheric conditions at below-, near- and above-rated wind speeds. The Kaimal and Mann models were fitted to these LES-generated turbulent inflows. Additionally, a constrained turbulence generator, PyConTurb [33], was used to achieve higher resolution compared to the saved output from LES. Waves were included, but, as shown by Torsvik et al. [23], the effects of waves on bearing loads of a bottom-fixed turbine are likely limited.

In Section 4, the dynamic analysis results and basic rating life estimates are presented for the DTU 10-MW RWT with simply supported main bearings. The characteristics of the obtained turbulent wind fields are first documented. Then, main bearing fatigue estimates are presented, and variations in results among the turbulence models are discussed. Section 5 summarizes conclusions and presents future work.

2 | Drivetrain Model

2.1 | The DTU 10-MW Reference Turbine

The new model is implemented for the DTU 10-MW reference turbine [31], with a medium-speed drivetrain developed at the Norwegian University of Science and Technology [32]. This turbine is an upwind turbine with a variable-speed collective-pitch controller. Table 1 gives an overview of the main turbine

parameters. For the first part of this work, which focuses on the implementation of the drivetrain model, the land-based version of the turbine is considered.

The drivetrain developed by Wang et al. [32] is a medium-speed, four-point support drivetrain, shown in Figure 1. It has two main bearings, both double-row tapered roller bearings taking up axial and radial loads, and moments, and two gearbox torque arm supports which transfer torsion to the bedplate.

2.2 | Simplified Drivetrain Model

The basis for the simplified drivetrain model is presented in Figure 2. Springs represent the stiffness of the main bearings ($\mathbf{K}_{MB1}$, $\mathbf{K}_{MB2}$) and torque arms ($\mathbf{K}_{TA}$). The shaft and bedplate can be modeled as flexible beams. In this work, however, the bedplate is rigid, a choice further discussed in Section 2.4.3. The gearbox and planet carrier are modeled as a common point mass, $\mathbf{M}_{GB}$, connected to the shaft by a rigid link and to the bedplate through two torque arms separated laterally. This simplification of the gearbox is counter to recommendations by Guo et al. [34] and was made for the sake of simplicity and because gearbox component loads are not considered here. Modeling the gearbox and planet carrier stiffness by means of springs is a straightforward future extension but requires appropriate input parameters. Similarly, the generator mass is modeled as a point mass,

$\mathbf{M}_{Gen}$, connected to the bedplate by a rigid link. This is equivalent to assuming that the generator coupling does not carry loads or moments other than torque, consistent with assumptions from previous work [25, 26, 28]. This representation was used as a basis for the implementation in the aero-hydro-servo-elastic code, leading to a model that will be referred to as an “extended global model”. Note that, in this work, the drivetrain is modeled both by a rotating and a nonrotating system. Figure 2 represents the nonrotating model. The rotating shaft and generator are modeled in the rotating system. More information on the separation into two systems and the modeling of the drivetrain components is given in the following sections.

2.3 | Implementation in an Aero-Hydro-Servo-Elastic Code

2.3.1 | OpenFAST

OpenFAST is an aero-hydro-servo-elastic engineering tool for the time-domain simulation of wind turbines. It uses a modular framework. Traditionally, a module called ElastoDyn is

TABLE 1 | DTU 10-MW reference wind turbine parameters.

Parameter	Value	Unit
Hub height	119	m
Rotor diameter	178.3	m
Rated wind speed	11.4	m/s
Rated power	10	MW
Rated rotor speed	9.6	rpm
Hub overhang	7.1	m
Shaft tilt angle	5.0	°
Rotor mass	227962	kg
Nacelle mass	446036	kg
Gearbox ratio	50	—

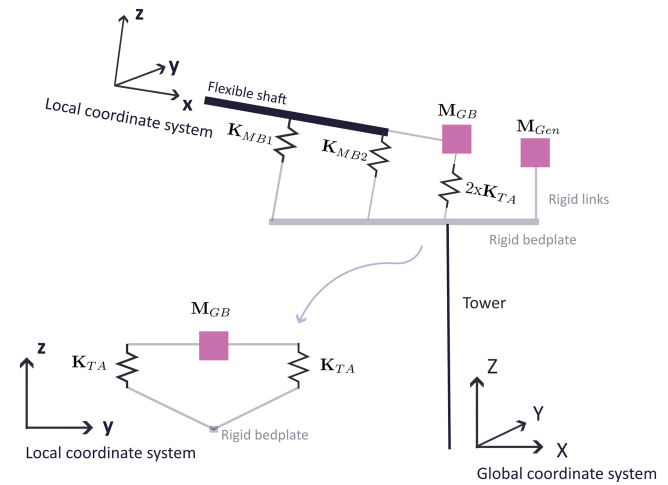


FIGURE 2 | Simplified, nonrotating drivetrain model and utilized coordinate systems. $\mathbf{K}_{MB1}$ and $\mathbf{K}_{MB2}$ are the stiffness matrices of the up-wind and downwind main bearings, respectively. $\mathbf{K}_{TA}$ is the stiffness matrix of each torque arm. For visualization purposes, the stiffness matrices are here represented by one-degree-of-freedom springs. The complete stiffness matrices applied for the drivetrain in this work are listed in Tables 2 and 6. $\mathbf{M}_{GB}$ is the gearbox and planet carrier mass. $\mathbf{M}_{Gen}$ is the generator mass.

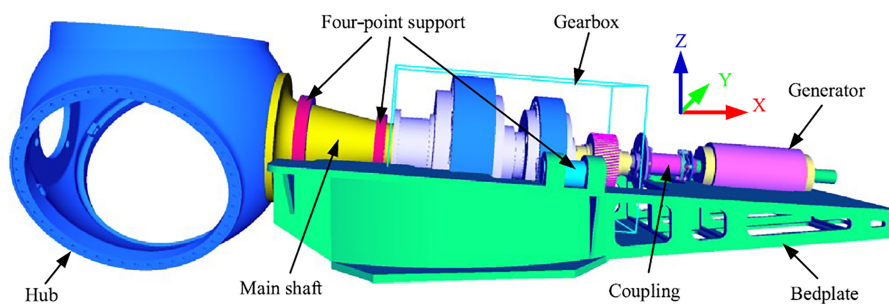


FIGURE 1 | Medium-speed drivetrain [32].

used for modeling the rotor, drivetrain, nacelle and tower in OpenFAST. It applies a modal representation of the blades and tower together with multibody representation of the nacelle and hub. The drivetrain is represented by means of a torsional linear spring-and-damper model in the low-speed shaft. Another module, SubDyn, was developed for modeling slender substructures such as jackets. It uses a linear finite-element beam framework, together with a Craig-Bampton [35] dynamic system reduction and a static improvement method. This allows for a large reduction in the number of modes while retaining an accurate solution. The flexibility of the finite-element framework facilitates unconventional geometries.

2.3.2 | Implementation in OpenFAST

Because adding degrees of freedom to ElastoDyn would be complicated, SubDyn emerges as a more adaptable and powerful option. SubDyn can model turbine towers up to the yaw bearing, allowing the inclusion of tower torsional flexibility and more than four tower bending modes (limits of ElastoDyn). However, the structural model in SubDyn is purely linear (necessitating small deformations), whereas ElastoDyn, which accounts for geometric nonlinearities in tower deflection, is more accurate for tubular towers [36]. Until recently, SubDyn had not been used in the context of modeling drivetrains. Parallel to the development of the presented research, however, a novel turbine concept was investigated that also uses SubDyn to model parts of a nonconventional drivetrain [37].

An obvious drawback of SubDyn's capability to model drivetrain components is its assumption of small-angle rotations. A drivetrain inevitably contains rotating parts. The shaft rotation about its longitudinal axis is considered by partially modeling it in a rotating framework and partially in a nonrotating framework, as shown in Figure 3. The rotating part applies the traditional way of modeling the drivetrain and rotor in OpenFAST, using the ElastoDyn module. It accounts for torsional stiffness of the shaft, gearbox, and generator in the traditional way: by means of an equivalent torsional spring. The drivetrain torsional spring stiffness is set to 2.317E9 Nm/rad, and the drivetrain torsional damper has

a value of 8.95E6 Nm/(rad/s). Rotational inertia of the generator, shaft, and rotor must be included on the rotating side. This is also where the generator torque is applied. Note that the nacelle yaw degree of freedom must be disabled with the current setup.

The nonrotating model in SubDyn includes the tower, the bedplate, the gearbox and generator mass, a flexible shaft, main bearings, and torque arms. The flexible shaft is modeled using linear Euler-Bernoulli beam elements and is stiff in torsion, because torsional flexibility is accounted for on the rotating side in the traditional manner using ElastoDyn. In drivetrain models, the main bearings and torque arms are typically simplified as springs. A spring element was implemented in SubDyn (distinct from beam elements previously available), significantly improving OpenFAST's ability to model a drivetrain. The spring element is massless, has zero length, and is specified by means of a symmetric 6-by-6 stiffness matrix. Modeling the complexly shaped drivetrain components as spring and beam elements is a challenge. A description on pitfalls and considerations when modeling a drivetrain in OpenFAST is provided by Krathe [38]. For the land-based turbine, 12 Craig-Bampton modes were necessary to accurately capture shaft and bearing loads. Rayleigh mass- and stiffness-proportional damping of the Guyan interface modes was governing for the extended global model, with mass and stiffness proportional coefficients of 0.0129 and 0.0004, respectively. The damping of the Craig-Bampton modes was set to 1% of critical.

2.4 | Drivetrain Components

2.4.1 | Coordinate System

A common coordinate system is used for all the flexible components of the drivetrain. The local and global coordinate systems are illustrated in Figure 2. The local x-axis is parallel to the shaft and positive downwind. The z-axis points (nominally) upwards, and the y-axis transversely and parallel to the global Y-axis. For a tilted shaft (5° for this turbine), the x- and z-axes are not completely aligned with the global axes. α , β , and γ denote rotation about x, y, and z, respectively. The z-axis is sometimes termed "vertical", while "axial" and "transverse" refer to the x-axis and y-axis, respectively. Following the coordinate convention, bearing radial loads are found as $F_r = \sqrt{F_y^2 + F_z^2}$.

2.4.2 | Main Bearings and Torque Arms

Simple (constant stiffness and uncoupled) translational and rotational springs are used to represent the main bearings and torque arms. The main bearings have zero stiffness in torsion, and the remaining diagonal terms are nonzero, because double-row tapered roller bearings support axial loads, radial loads, and moments. The torque arms have nonzero elements on the entire diagonal of their stiffness matrices. The bearing and torque arm stiffness matrices are taken from the medium-speed drivetrain modeled by Wang [32], where the stiffness matrix off-diagonal terms were assumed to be zero. However, it is fully possible to include off-diagonal terms. The applied stiffnesses are summarized in Table 2, in accordance with the local coordinate system in Figure 2. The torque arms, located on each side of the

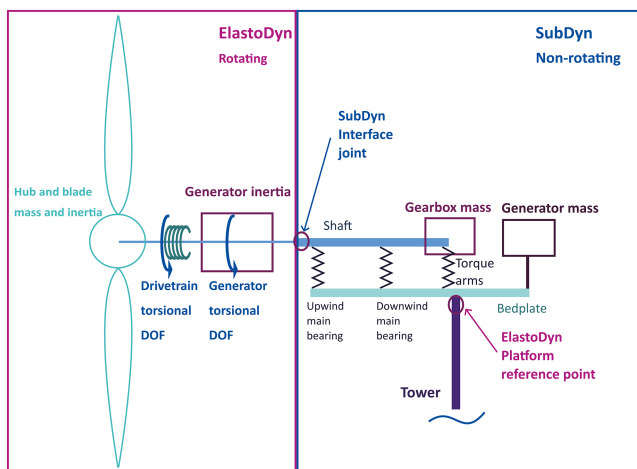


FIGURE 3 | OpenFAST drivetrain model.

TABLE 2 | Bearing and torque arm contributions to the stiffness matrix in the local coordinate system.

Component	K_{xx} (N/m)	K_{yy} (N/m)	K_{zz} (N/m)	$K_{\alpha\alpha}$ (Nm/rad)	$K_{\beta\beta}$ (Nm/rad)	$K_{\gamma\gamma}$ (Nm/rad)
Upwind main bearing	3.53E+09	8.92E+09	1.25E+10	0	1.21E+09	8.62E+08
Downwind main bearing	3.39E+09	5.38E+09	8.78E+09	0	5.92E+08	3.62E+08
Torque arms	1.20E+08	2.40E+08	2.40E+10	1.20E+08	2.40E+08	2.40E+08

gearbox, take up drivetrain torque. This is manifested by opposing vertical forces on the left and right support. Consequently, in the drivetrain model, torque arm springs are laterally offset with rigid links (see Figure 2). Both main bearing springs and torque arm springs are located between two joints along the shaft. For each component, one joint is connected to the shaft, and one is connected to a rigid link extending to the bedplate.

2.4.3 | Shaft and Bedplate

The shaft is modeled by means of a flexible beam. The dimensions and material properties of the shaft are obtained from the reduced-order linear flexible shaft in the multibody drivetrain model developed by Wang [32]. By applying static forces and moments to the shaft tip, fixing the shaft on the downstream side, and extracting corresponding displacements and rotations, shaft material and geometric properties were estimated.

The bedplate is modeled here as a rigid beam. Wang et al. [21] recommended modeling the bedplate as a flexible structure to properly represent rotor-drivetrain-tower coupling and obtain more accurate bearing fatigue damage. This is in agreement with the authors' expectations for large wind turbines. However, the bedplate designed for the DTU reference turbine is particularly difficult to model by means of equivalent beams because of its complex geometry. Omitting the flexibility of the bedplate here does not mean that it is not important but rather that modeling this specific bedplate by means of equivalent beams would not give a satisfactory result.

2.5 | Verification

2.5.1 | Coupled Multibody Model

A fully coupled model of the land-based DTU 10-MW wind turbine, including a detailed drivetrain, is used as a means to verify the drivetrain implementation in OpenFAST. The model is built in the multibody simulation software Simpack [39]. The drivetrain model was developed by Wang et al. [32]. The bedplate is rigid, the shaft is flexible, and the main bearings and gearbox torque arms are modeled as translational and rotational uncoupled springs with constant stiffness. Unlike the extended global model, this more complete multibody global model includes gears, flexible high-speed shafts, a flexible planet carrier, and a flexible generator coupling. Flexible tower and blades were modeled by Monnac [40]. These two models were compiled into a single coupled model. The implemented drivetrain model in OpenFAST is referred to as the “extended global model”, whereas the conventional modeling approach, where everything

is modeled in ElastoDyn, is referred to as the “traditional global model”. The coupled multibody model in Simpack will be referred to as the “multibody global model” or “multibody model”. System properties and global behavior have been verified against the traditional DTU 10-MW turbine in OpenFAST [41]. The detailed drivetrain model is used to obtain input to the extended global model, as described previously. Then, the response and behavior of the multibody model are used to verify the implementation.

The multibody coupled model is computationally expensive. When comparing the simulation time of the extended global model and the multibody model, the extended global model is in the order of 500 times faster, with 700 s of simulations taking several days for the multibody model. The main reason for the slow simulations of the multibody model is that it captures details of the high-speed rotating components at high (i.e., gear mesh) frequencies. The new tight coupling implementation forthcoming in OpenFAST v5.0 [42] is expected to further increase the speed by a factor of 10 to 100, making the extended global model a very attractive option, even if the multibody model had been simplified in order to reduce computational costs.

2.5.2 | Modal Analysis

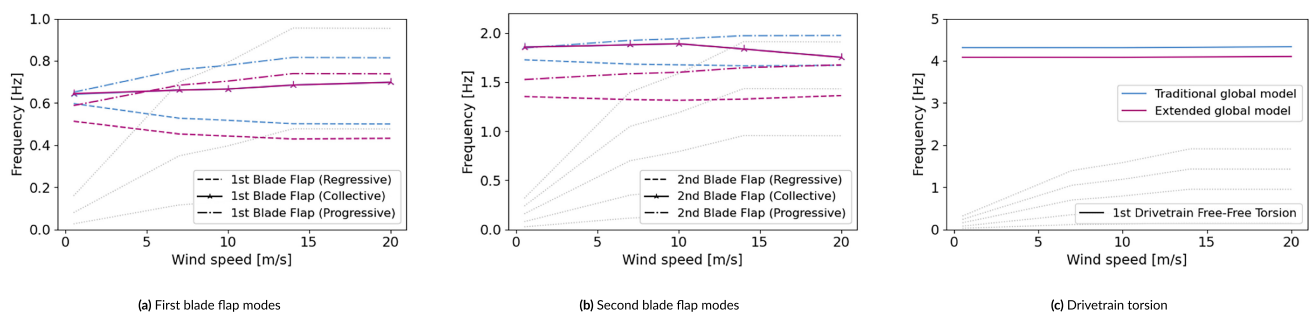
2.5.2.1 | Frequencies. Undamped natural frequencies of the traditional, extended, and multibody global models are presented in Table 3. The values were obtained through modal analysis of a standstill turbine.

Table 3 shows that the multibody model's drivetrain torsional frequency (mode 3) is lower compared to the two other models, attributed to its many sources of torsional flexibility. The slight difference in torsional frequency for the traditional and extended global models is due to torsional flexibility from the torque arms in the drivetrain model. More significant differences are seen in the asymmetric flap modes. The natural frequencies of flap modes with yaw are higher in the traditional model, partly due to coupling with tower torsion in the drivetrain models (the traditional global model does not include tower torsion). Drivetrain flexibility causes both flap modes to occur at lower natural frequencies. Discrepancies in the rotor edge mode frequencies also appear between the multibody model and the other models, likely due to the multibody drivetrain being softer in torsion.

Linearizations at different wind speeds were performed for the traditional and extended global models but were not possible to obtain for the multibody model. The tower mode frequencies vary little with wind speed. Moreover, there were no significant differences between the models considering edgewise bending

TABLE 3 | Undamped natural frequencies of standstill land-based turbine.

Mode	Description	Undamped natural frequency (Hz)		
		Traditional global model	Extended global model	Multibody global model
1	First tower side-side mode	0.25	0.25	0.24
2	First tower fore-aft mode	0.25	0.25	0.25
3	First fixed-free drivetrain mode	0.53	0.51	0.46
4	First asymmetric flap with yaw	0.63	0.53	0.52
5	First asymmetric flap with tilt	0.62	0.57	0.56
6	First collective flap mode	0.64	0.64	0.62
7	First asymmetric edge mode 1	0.99	0.98	0.92
8	First asymmetric edge mode 2	1.0	0.99	0.93

**FIGURE 4** | Campbell diagrams showing natural frequencies of the traditional and global models. Dashed lines represent 1P, 3P, and 6P from bottom to top. For (b) and (c), 9P and 12P are also included. The collective flap modes of the traditional and extended models coincide.

of the blades. These results are therefore omitted here. Campbell diagrams of blade flap modes and drivetrain torsion are shown in Figure 4a–c.

Consistent with Table 3, the main takeaways from Figure 4a–c are as follows:

- The asymmetric blade flap modes have reduced natural frequencies for the extended global model.
- The drivetrain torsional mode in the extended global model occurs at reduced frequencies also when the turbine is rotating.

2.5.2.2 | Damping. Structural damping for different turbine modes was obtained similarly as for the natural frequencies and is presented in Table 4. The structural damping of the first eight modes mostly matches well between the models. However, the traditional model has higher damping of the asymmetric flap modes than the extended model and the multibody model. In the extended model, a single damping quantity is assigned, corresponding to the damping of the flap mode of a *single* blade. ElastoDyn uses stiffness-proportional damping to extrapolate to frequencies other than the natural frequency of this mode. As seen previously, the natural frequencies of the asymmetric flap modes differ for the different models. Using stiffness-proportional damping, it is not possible to

tune the damping to simultaneously obtain similar damping of the asymmetric flap modes *and* the collective flap mode for all models.

2.5.3 | Structural Response to Turbulent Wind

Three different cases of turbulent wind were run for the three global models to compare load responses in the drivetrain. After discarding 100 s of initial transients, each design load case (DLC) had a duration of 600 s, corresponding to 10 min of stationary wind. The turbulent wind was generated using the Kaimal spectrum with exponential coherence model, with turbulence parameters based on wind turbine class A and a power law shear profile. The tower influence on the aerodynamic loads was not included. Main characteristics of the DLCs are shown in Table 5. For the sake of conciseness, the results of only one load case, DLC15, are presented. The conclusions from the other DLCs are similar.

Time series of the main bearing loads are presented in Figure 5, and the response amplitude spectra of the bearing loads are presented in Figure 6. The dynamics of the main bearing loads are generally well captured by the extended global model. The multibody simulation shows a wide peak at 4 Hz, which is not found for the extended model. The reason for this discrepancy is not known.

TABLE 4 | Structural damping of the land-based turbine in a standstill condition.

Mode	Description	Critical damping ratio (%)		
		Traditional global model	Extended global model	Multibody global model
1	First tower side-side mode	0.30	0.29	0.30
2	First tower fore-aft mode	0.31	0.30	0.31
3	First fixed-free drivetrain mode	0.42	0.43	0.42
4	First asymmetric flap with yaw	0.48	0.33	0.29
5	First asymmetric flap with tilt	0.47	0.38	0.38
6	First collective flap mode	0.50	0.50	0.50
7	First asymmetric edge mode 1	0.47	0.47	0.47
8	First asymmetric edge mode 2	0.48	0.48	0.48

TABLE 5 | Turbulent wind load cases for model verification.

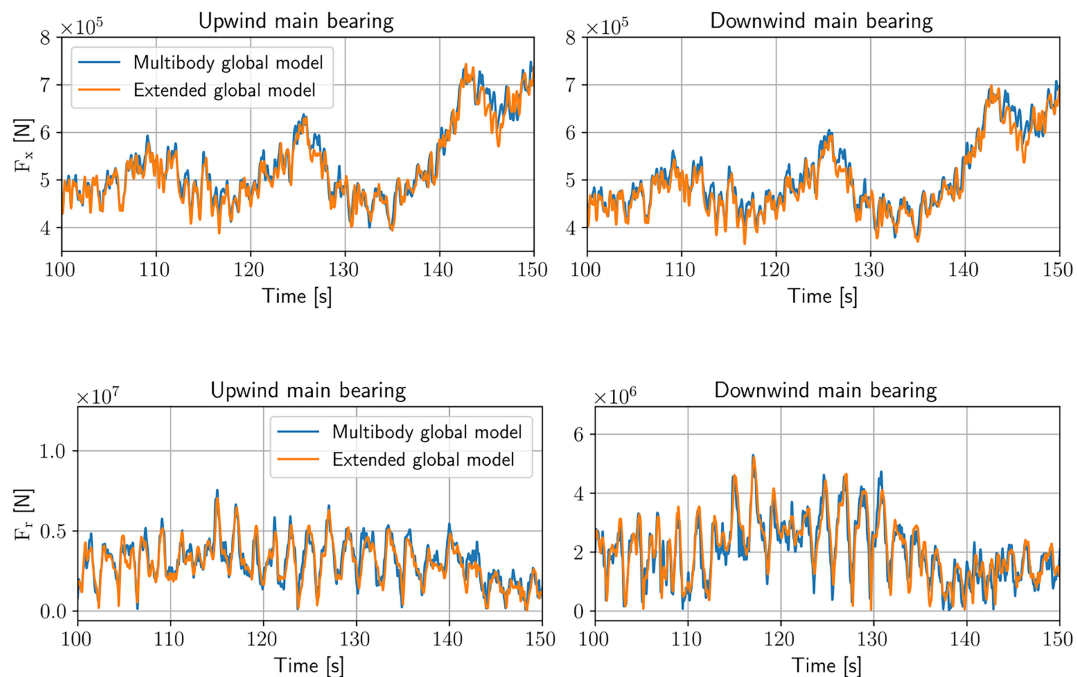
Load case label	Mean wind speed at hub height (m/s)	Turbulence intensity at hub height (%)	Shear exponent [-]
DLC7	7	24.8	0.2
DLC10	10	21.0	0.2
DLC15	15	18.0	0.2

3 | Dynamic Analyses

This section presents the dynamic analyses applied to investigate bearing fatigue sensitivity to different turbulence models. These analyses were facilitated by the computational efficiency of the new extended global model.

3.1 | Base Case and Numerical Setup

For this study, an offshore, bottom-fixed version of the DTU 10-MW RWT was utilized, applying the monopile developed in OC 6 phase II [43] with a flexible foundation at 30-m

**FIGURE 5** | Time series of main bearing axial (top) and radial (bottom) forces for DLC15.

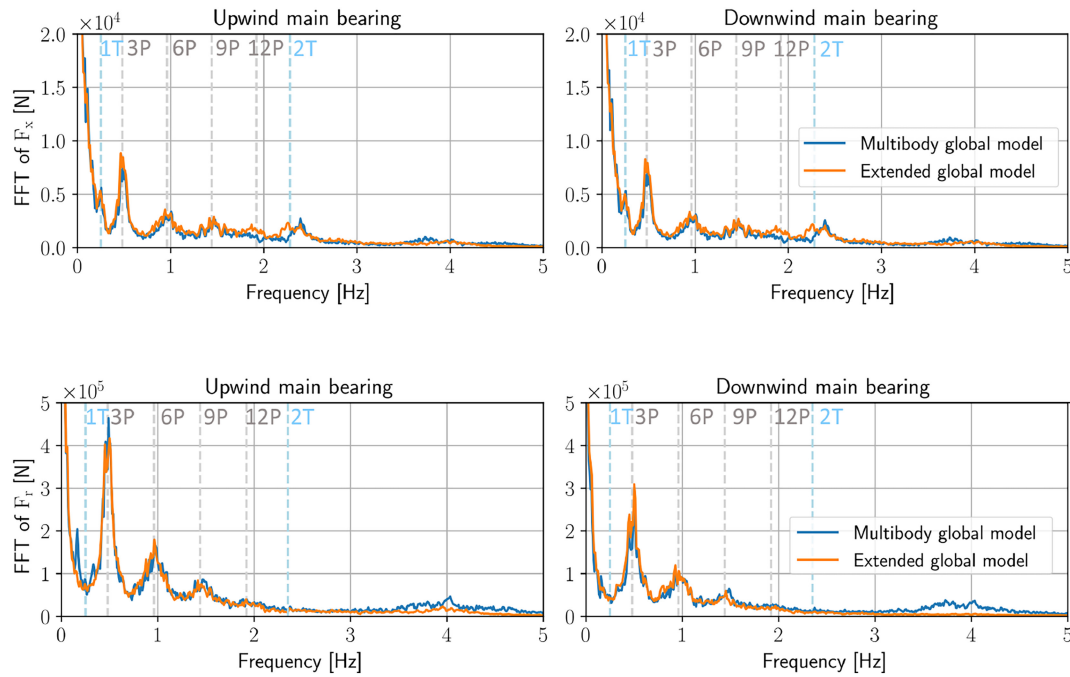


FIGURE 6 | Fast Fourier transform of main bearing axial (top) and radial (bottom) forces. The gray dashed lines represent (from left to right) 3P, 6P, 9P, and 12P. The light blue dashed lines represent the first (1T) and second (2T) tower fore-aft bending modes of the extended global model.

TABLE 6 | Bearing contributions to the stiffness matrix in the local coordinate system for the investigation of main bearing fatigue sensitivity to turbulence models.

Component	K_{xx} [N/m]	K_{yy} [N/m]	K_{zz} [N/m]	$K_{\alpha\alpha}$ [Nm/rad]	$K_{\beta\beta}$ [Nm/rad]	$K_{\gamma\gamma}$ [Nm/rad]
Upwind main bearing	3.53E+09	8.92E+09	1.25E+10	0	0	0
Downwind main bearing	3.39E+09	5.38E+09	8.78E+09	0	0	0

water depth. The drivetrain implementation presented and verified in Section 2 was applied. However, to obtain accurate fatigue damage of bearings subjected to bending moments, detailed fatigue analyses are necessary, as described in ISO/TS 16281 [44] and exemplified by Stirling [45]. The information required to perform such analyses was not available for the bearings in question. To ease fatigue calculations, main bearings without moment stiffness were considered for the remaining work, and their stiffness matrices are presented in Table 6. Blade pitch and generator torque control were included using the ROSCO controller [46]. The tower influence model (tower shadow) considered a rigid tower extending to the ground from the tower top position. That is, the influence of tower elastic deflection on tower shadow was not included. The tower influence model used potential flow theory with the Bak correction [47].

Dynamic simulations were run with an OpenFAST glue-code time step of 0.01 s. To appropriately solve axial loads in the shaft and bearings, 14 Craig–Bampton modes were included in SubDyn, corresponding to natural frequencies up to 17.6 Hz. The fourth-order Adams–Bashforth integrator was applied, requiring a time step of 0.0025 s in SubDyn. A correction iteration was utilized in the OpenFAST glue-code (recommended

when SubDyn is used). In total, this resulted in a 60% increase in computation time compared to a traditional OpenFAST simulation.

3.2 | Environmental Load Cases

Three different mean wind speeds at hub height, U_{hub} , were considered, representing the operational regions of the turbine. For each wind speed, three different stability conditions of the atmosphere were used, with synthetic turbulent wind fields generated by three turbulence generators. The significant wave height and peak period, H_s and T_p , were based on 3 years of reanalysis from ERA5 [48] at the FINO1 [49] location. The 90th percentile of H_s conditional on the wind speed was used, together with the mean T_p conditional on this H_s . Irregular waves were generated from the Pierson–Moskowitz spectrum [50]. The environmental load cases are presented in Table 7.

3.3 | Synthetic Turbulence Models

For each combination of mean wind speed and wave conditions, stable, neutral, and unstable atmospheric conditions were

TABLE 7 | Environmental load conditions.

EC	U_{hub} (m/s)	H_s (m)	T_p (s)	Atmospheric conditions	Synthetic turbulence generators
EC1	7.5	1.8	7.1	Stable	Mann
EC2	12.0	2.6	7.7	neutral	Kaimal
EC3	16.0	3.6	8.7	unstable	PyConTurb

considered. Holtslag [51] provides a thorough explanation of stable, neutral and unstable atmospheric conditions. In short: turbulence intensity increases as stability decreases (stable to neutral to unstable), while shear exhibits the opposite trend.

Three methods for generating synthetic turbulent wind fields were applied for each atmospheric condition, namely, the Mann spectral tensor model [3, 4] (Mann), the Kaimal spectrum, and exponential coherence model [2] (Kaimal) and PyConTurb [33]. The two former were generated using the IEC Turbulence Simulator [52] and TurbSim [53], respectively. The latter is based on the Kaimal model but generates constrained turbulence based on point measurements, aiming to recreate correct spatial coherence.

The synthetic wind fields were fitted to LES generated for a characteristic location in the North Sea [49] using the Weather Research and Forecasting model [54]. The fitting of turbulence model parameters is elaborated in Sections 3.3.1–3.3.3. Details of the LES can be found in the work of Rivera et al. [7] To obtain statistical confidence in the turbine structural response, six 1-h realizations were applied. For PyConTurb, some conditions made use of four or five seeds. The realizations were obtained from LES by storing wind fields from six neighboring locations in the simulated space. The reason for not applying LES wind fields directly in the simulations is discussed in Section 3.3.3. Table 8 provides details of the numerical setup.

The longitudinal discretization, Δx , relates to the duration of the turbulence simulation, T , the number of points in the longitudinal direction, N_x , and the mean wind speed at hub height, as

$$\Delta x = \frac{TU_{hub}}{N_x} \quad (1)$$

and the time step is

$$\Delta t = \frac{T}{N_x}. \quad (2)$$

3.3.1 | Kaimal

In the Kaimal model, the turbulence is generated based on the Kaimal spectrum:

$$\frac{fS_i(f)}{\sigma_i^2} = 4 \frac{\frac{fL_i}{U_{hub}}}{\left(1 + 6 \frac{fL_i}{U_{hub}}\right)^{5/3}}, \quad (3)$$

where $i = u, v, w$ refers to the velocity component of the wind, L_i is the integral scale, f is the frequency in hertz, $S_i(f)$ is the one-sided spectrum, and σ_i is the standard deviation of the wind velocity component in question.

Exponential coherence is imposed on the generated turbulence as follows:

$$\gamma_{ii}(\delta, f) = \left[\exp \left(-a_i \sqrt{\left(\frac{f\delta}{U_{hub}} \right)^2 + (b_i\delta)^2} \right) \right]^2. \quad (4)$$

Here, δ is the distance between two points in the vertical two-dimensional grid. The parameters a_i and b_i are the spatial coherence decrement and offset parameter of each wind component. This exponential coherence is applied for all three components of the wind but only for coherence between the same components and only for points separated in the vertical and/or cross-wind direction. The parameters a_i and b_i were fitted to LES data using four pairs of points, centered at the hub and evenly distributed around it, each separated by a distance of $\delta = 0.63D$, with a least squares fit of the co-coherence spectra at 0 to 0.05 Hz.

3.3.2 | Mann

The Mann model was developed for neutral atmospheric conditions and calibrated against measurements over a bridge. It is based on rapid distortion theory, where isotropic turbulence is distorted by uniform vertical shear. The Mann model, when applying parameters according to the IEC standard, uses the same spectrum as the Kaimal model but applies a three-dimensional spectral tensor, and the three turbulence components (u, v, w) are generated in a single three-dimensional fast Fourier transform, implying dependent components that can more realistically represent the correlation between along-wind and cross-wind components [55].

The Mann model attempts to describe the turbulent wind field by three input parameters, $\alpha\epsilon^{2/3}$, L and Γ . The former describes the dissipation of turbulent kinetic energy, whereas L is the length scale of the velocity spectral tensor. The latter, Γ , is the shear distortion parameter, related to the lifetime of the turbulent eddies. These parameters were fitted to LES data at a height of 131.7 m by Rivera et al. [7], applying tools made by Cheynet [56]. This algorithm simultaneously fits the three parameters to the three wind velocity component spectra and to the real part of the cross-spectrum of u and w , by means of a nonlinear least squares fit.

Coherence in the Mann model can be expressed in terms of its spectral velocity tensor, ϕ_{ij} , and is a function of the wave number (a spatial form of frequency) in the along-wind direction, k_1 , and the spatial separations of the points, δy and δz :

$$\gamma_{ij}(k_1, \delta y, \delta z) = \frac{\chi_{ij}(k_1, \delta y, \delta z)}{\sqrt{F_i(k_1)F_j(k_1)}}, \quad (5)$$

where

TABLE 8 | Numerical setup of turbulent wind fields.

Turbulence generator	Δt (s)	$\Delta y, \Delta z$ (m)	Δx (m) (EC1, EC2, EC3)	Width (m)	Height (m)
LES	0.1	8	0.75, 1.20, 1.60	224	232
PyConTurb	0.1	4	0.75, 1.20, 1.60	200	200
Mann	0.06	4.02	0.46, 0.73, 0.98	257.41	257.41
Kaimal	0.06	4.02	0.46, 0.73, 0.98	249.36	249.36

$$\chi_{ij}(k_1, \delta y, \delta z) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \phi_{ij}(k_1, k_2, k_3) e^{-ik_2 \delta y} e^{-ik_3 \delta z} dk_2 dk_3 \quad (6)$$

and

$$F_i(k_1) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \phi_{ii}(k_1, k_2, k_3) dk_2 dk_3. \quad (7)$$

The wave number in the along-wind direction, k_1 , is expressed in terms of frequency as $k_1 = \frac{2\pi f}{U_{hub}}$, assuming frozen turbulence.

Moreover, $i, j = \{u, v, w\}$, as before, and k_2 and k_3 are the wave numbers in the y - and z -direction.

3.3.3 | PyConTurb

The LES data from Rivera et al. [7] were extracted at a span-wise resolution of 8 m. This might be sufficient for the motion response of a large floating turbine, but a grid resolution close to the mean chord length is recommended for calculating structural responses of elastic components [53]. This corresponds to approximately 4 m for the DTU 10-MW RWT. A simple check with turbulent wind generated using the Kaimal model (not presented here) showed that the peak of the bearing response spectrum at 3P could increase by a factor of 2 by going from 8-m to 4-m grid resolution. Therefore, it was deemed necessary to apply a spatial resolution of 4 m when estimating bearing fatigue.

It is trivial to regenerate synthetic turbulence with increased grid resolution. However, it was not practically possible to generate new LES data. An appealing solution would be to interpolate wind velocities between neighboring grid points. Yet, doing so would merely replicate what the aerodynamic load calculations are already doing, without properly capturing the temporal variation in the blade and shaft loading due to spatial variation in the turbulence grid.

To solve this challenge, PyConTurb [33] was applied. It is an open-source constrained-turbulence generator originally developed to generate synthetic turbulence based on observed data. PyConTurb aims at reconstructing spatial coherence between simulated points and all the user-provided time series (*constraints*). The time series of the constraints, the LES data, are kept. Simulated time series are added to points between the constraints. PyConTurb was set to linearly interpolate turbulence power spectra, $S_i(f)$, standard deviation, σ_i , and mean

TABLE 9 | Number of realizations and ratio of LES grid points used as constraints for PyConTurb.

Condition	U_{hub}		
	7.5 m/s	12 m/s	16 m/s
Stable	4 × 100%	5 × 100% + 1 × 67%	4 × 100% + 2 × 67%
Neutral	6 × 100%	3 × 100% + 2 × 80%	6 × 100%
Unstable	5 × 100%	3 × 100% + 3 × 80%	4 × 100% + 1 × 80%

wind speed for simulated points, for all components $i = \{u, v, w\}$. Additionally, correlation between the simulated and constrained points is ensured using the procedure described by Rinker [33]. PyConTurb applies the exponential coherence model for all three wind components with input parameters based on the recommendations by IEC [1]. Likely due to an overconstrained problem, some realizations of the LES data were not possible to reproduce sufficiently accurately using PyConTurb. Some conditions therefore have fewer than six seeds. For the same reason, in some cases, only 2/3 or 4/5 of the LES grid points were used to constrain PyConTurb. This is summarized in Table 9. To the authors' knowledge, PyConTurb has not previously been applied for the purpose of increasing resolution of already generated wind fields. Figure 7 illustrates the ability of PyConTurb to capture the larger structures of LES simulations while adding grid points to increase resolution.

4 | Results

4.1 | Turbulence

A comparison of the characteristics of the different turbulent wind fields is presented. The LES data were not directly applied in the dynamic simulations with a wind turbine but are included here as a baseline. The vertical shear profile and turbulence intensity of the different wind fields are presented in Figure 8. Unstable conditions are characterized by higher turbulence intensity than neutral and stable conditions, and they exhibit very little shear.

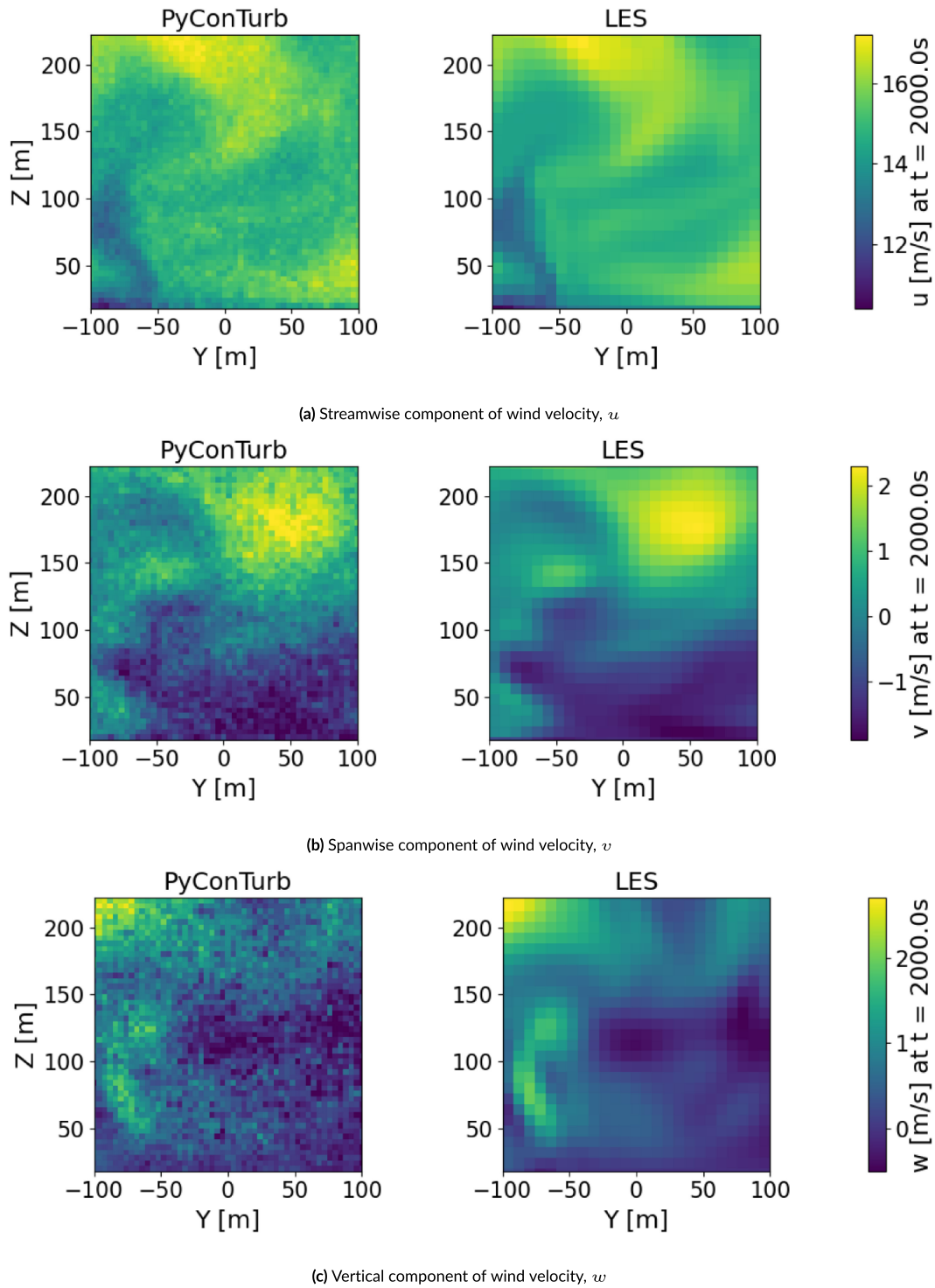


FIGURE 7 | Comparison of instantaneous wind velocity of PyConTurb generated based on LES ($\Delta y, \Delta z = 4$ m) and LES ($\Delta y, \Delta z = 8$ m).

The power spectral densities of the u -component of the wind fields at the hub are presented in Figure 9 for the 7.5 m/s condition and in Appendix A for 12 and 16 m/s. The spectra were obtained using Welch's method with Hamming window using six segments and 50% overlap. The same procedure was applied for other spectra discussed in following sections. For PyConTurb, time series 4m next to the hub were applied to avoid taking points that were identical to the LES results.

LES resolves only the large eddies of the turbulent flow. For this reason, and previously noted by many [7, 12, 15, 57], LES are not able to resolve the higher frequencies of the inertial subrange that the Mann and Kaimal models exhibit (and which are present in wind measurements [57]) without significant increase in computational cost. This is seen for frequencies above approximately 0.1 Hz in Figure 9. Although high-frequency loads are expected to be important for drivetrain components, these

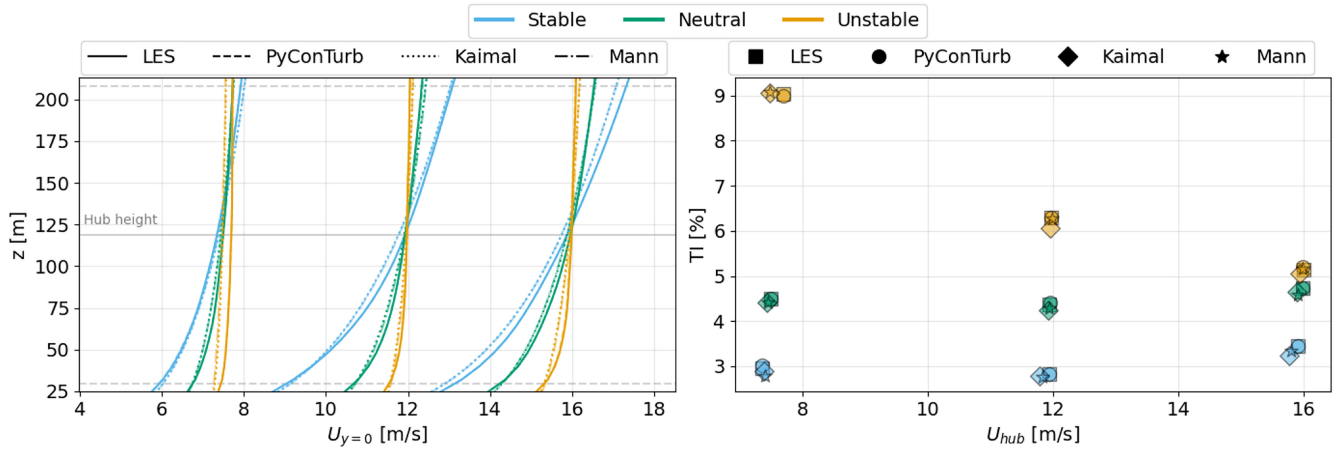


FIGURE 8 | Vertical shear profile at turbine center (left). Turbulence intensity plotted against mean wind speed at hub height (right). Both are averaged across seeds for each turbulence model and stability condition. The shear profiles from LES and PyConTurb coincide, as do the shear profiles of Kaimal and Mann.

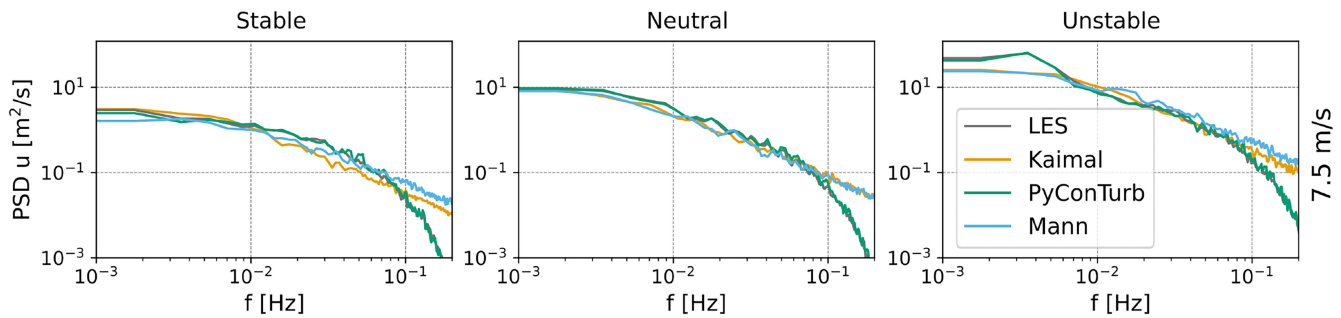


FIGURE 9 | Turbulence spectra of the longitudinal wind velocity component at hub height averaged over seeds for each turbulence model and atmospheric condition at 7.5 m/s.

loads are driven mainly by the blades passing through the varying wind field at high frequencies, not high-frequency content in the wind. As is evident from Figure 9, energy levels are low beyond 0.1 Hz. Applying a low-pass filter to the Kaimal wind fields (not shown here) showed that the high-frequency content had little influence on bearing fatigue, increasing it by 0.1%–3%. Therefore, the inability of LES to capture high-frequency energy content of the wind is not expected to significantly impact the fatigue comparisons.

At lower frequencies under unstable conditions, the LES spectrum has more energy than the Kaimal and Mann spectra, which are quite similar. Being designed for neutral conditions, Mann and Kaimal match the LES spectra for these conditions. In stable conditions, discrepancies occur between the models at low frequencies and in the inertial subrange. Taking LES (and PyConTurb) as the baseline, Kaimal overestimates low-frequency energy and underestimates inertial subrange energy. The Mann spectrum generally sits between LES and Kaimal but underestimates the low-frequency energy below rated.

4.1.1 | Spatial Coherence

Coherence provides information about correlation between time series of two spatially separated points in the frequency domain.

Nybø et al. [15] found that quad-coherence had negligible impacts on the DTU 10-MW bottom-fixed turbine's tower-top bending moments, and this assumption is here extended to shaft bending moments and bearing loads. This analysis is therefore confined to co-coherence, obtained by considering only the real value of the coherence, γ_{ij} :

$$\text{Re}(\gamma_{ij}(p_1, p_2, f)) = \frac{\text{Re}(S_{ij}(p_1, p_2, f))}{\sqrt{S_{ii}(p_1, f)S_{jj}(p_2, f)}} \quad (8)$$

Here, $i, j = \{u, v, w\}$ represent the along-wind, cross-wind and vertical turbulence components, and f is the frequency in hertz. $S_{ii}(p_1, f)$ and $S_{jj}(p_2, f)$ are the one-sided power spectra of the wind velocities at each respective point, p_1, p_2 , while $S_{ij}(p_1, p_2, f)$ denotes the cross-spectrum of these two time series. Focus is on the co-coherence of the same velocity component between two points separated in the plane normal to the mean wind direction: $i = j$, $p_1 = p(y_1, z_1)$ and $p_2 = p(y_2, z_2)$. Equation (8) was applied to evaluate the co-coherence of each turbulent wind field in question. The uu co-coherence will be considered, previously shown to be more important for tower loading in a bottom-fixed 5-MW turbine than vv and ww components [58]. This is also consistent with Lavelly [24], who found the spatial variations of the streamwise wind

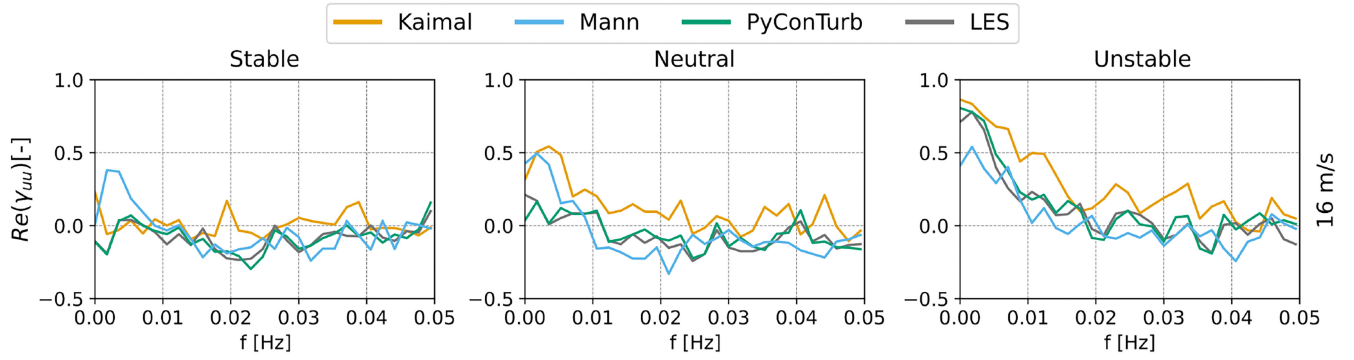


FIGURE 10 | Numerical co-coherence of the u -component for two points separated laterally and centered at the hub height at 16-m/s mean wind speed.

component most important for shaft out-of-plane bending moments and bearing forces.

Nybø et al. [15] revealed that high co-coherence (close to 1) at 0.7D separation reduced tower-top bending moments by causing more uniform rotor loading, whereas lower (but not very low) co-coherence increased them. Negative co-coherence also led to increased moments. Similarly, Putri et al. [59] found low lateral co-coherence to give higher tower-top torsional moments.

In Figure 10, the co-coherence is plotted against frequency for a lateral separation distance of about 0.7D, centered at the rotor, for the case with 16-m/s mean wind speed. Similar figures for all wind speeds, showing both lateral and vertical separation, are available in Appendix B. To assess PyConTurb's replication of LES, co-coherence was investigated at points shifted 4 m relative to the LES data. Clearly, PyConTurb matches the co-coherence of LES very well at the generated points.

In unstable conditions, Kaimal exhibits higher co-coherence than Mann and PyConTurb, except at very low frequencies. Mann shows lower lateral co-coherence than the others, and for the vertically separated points, this only applies below rated. In stable conditions, co-coherence levels are generally low, except for the Mann model, and for Kaimal's lateral co-coherence below rated. Under neutral conditions, LES has lower low-frequency co-coherence than both Kaimal and Mann.

There is overall more negative lateral co-coherence for PyConTurb and Mann. PyConTurb also exhibits some negative vertical co-coherence. Nybø's findings [15] suggest that negative lateral co-coherence can increase shaft yaw bending moments, further increasing transverse forces in the bearings.

4.1.2 | Asymmetry Index

Putri et al. [59] and Nybø et al. [15] noted the influence of an asymmetric longitudinal wind velocity distribution over the rotor on tower-top moments. Lavelly [24] introduced the triaxial asymmetry index, a measure of the asymmetric wind loading on the rotor, correlated with out-of-plane shaft bending moments and drivetrain failure.

The asymmetry index, Equation (9), considers asymmetric loading in both the horizontal and vertical directions and takes into account the blade sections contributing most to shaft loading.

$$I_A = \iint_A r(y, z) \delta u_T dA, \quad (9)$$

where

$$\delta u_T = \max \left(\left| u(r, \theta) - u \left(r, \theta + \frac{2\pi}{3} \right) \right|, \left| u(r, \theta) - u \left(r, \theta - \frac{2\pi}{3} \right) \right| \right) \quad (10)$$

is the maximum difference in the streamwise velocity, u , between three blade locations. A cylindrical coordinate system is applied, centered at the hub, where r is the radius and θ represents the vertical plane sweep angle. The result is a time-dependent parameter measuring the asymmetry of the wind field in the rotor plane. Examples of the instantaneous wind u -component corresponding to high and low asymmetry indices are shown in Figure 11. The power spectral density of the asymmetry index is presented together with main bearing load spectra in Figure 15 in Section 4.3.2 and in Appendix C. PyConTurb generally has more low-frequency variation in the asymmetry index, except at very low frequencies in stable conditions.

4.2 | Fatigue Calculations

As first presented by Palmgren in 1924 and according to ISO 281 [60], the bearing basic rating life is given as

$$L_{10} = \left(\frac{C_D}{P} \right)^p, \quad (11)$$

where L_{10} is in 10^6 revolutions and is the operational life which 90% of a population of bearings are expected to reach without failure by rolling contact fatigue. C_D is the dynamic load capacity, the load at which L_{10} equals 1 million revolutions. The exponent p is taken as 10/3 for roller bearings. P is the bearing dynamic equivalent radial load, calculated as follows:

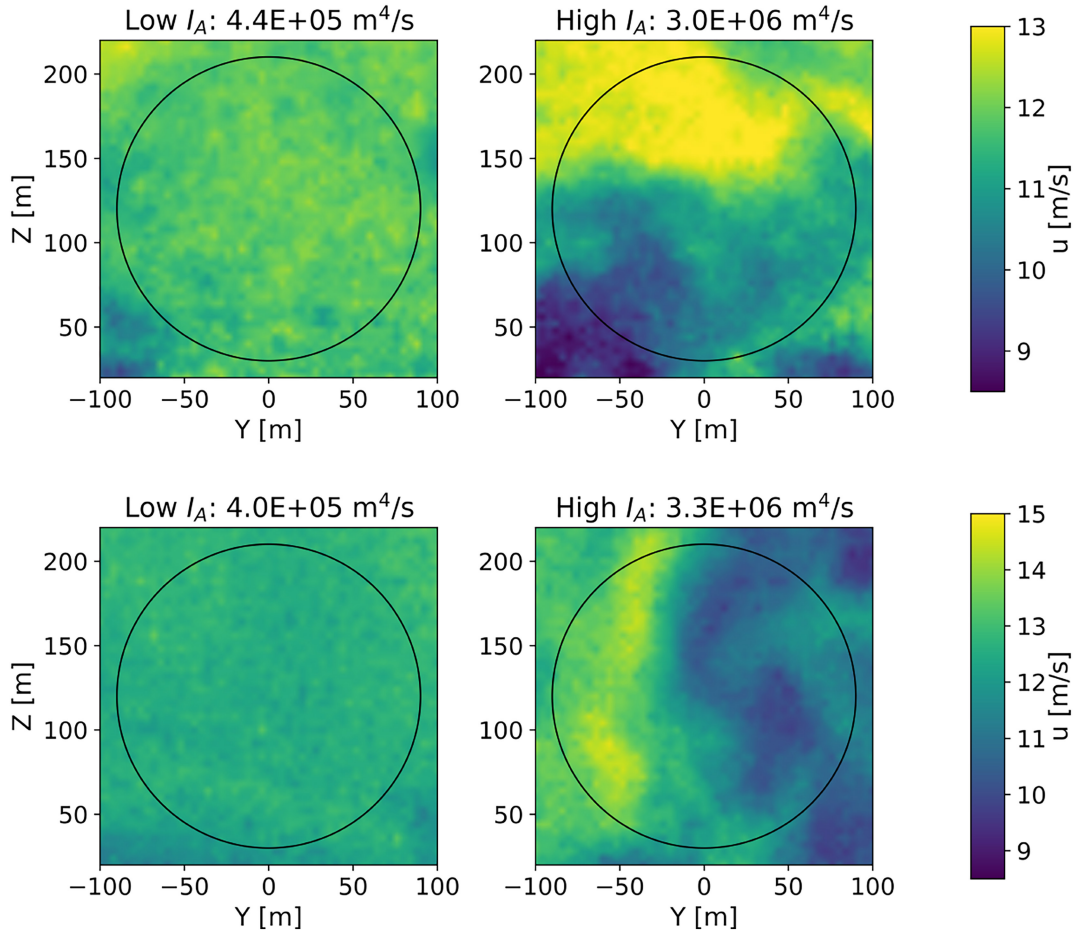


FIGURE 11 | Examples of wind field time instants with low and high asymmetry index, I_A .

$$P = XF_r + YF_a. \quad (12)$$

F_r is the bearing radial load, and F_a is the bearing axial load, $F_a = F_x$. X and Y are the radial and axial load factors, provided by the bearing manufacturer or estimated by ISO 281 [60] based on the bearing nominal contact angle. Their quantities depend on the ratio between the axial and radial bearing load and are provided in Table 10.

The main bearing fatigue can be calculated assuming linear cumulative damage, in accordance with the Palmgren–Miner rule:

$$D = \sum_i \frac{l_i}{L_i} = \frac{1}{10^6 C_D^p} \sum_i l_i P_i^p. \quad (13)$$

l_i is the number of revolutions at load i and can be found as

$$l_i = \frac{n_i t_i}{60}, \quad (14)$$

where n_i is the low-speed shaft rotational speed (rev/min) at load P_i and t_i is the time step. L_i is the L_{10} life at the bearing dynamic equivalent load range, P_i , in revolutions.

TABLE 10 | Main bearing fatigue parameters.

Bearing location	C_D (kN)	e (-)	X (-)	Y (-)	Load ratio
Upwind	9690	0.61	1	1.11	$\frac{F_a}{F_r} \leq e$
Main bearing			0.67	1.66	$\frac{F_a}{F_r} > e$
Downwind	7860	0.73	1	0.92	$\frac{F_a}{F_r} \leq e$
Main bearing			0.67	1.37	$\frac{F_a}{F_r} > e$

4.2.1 | Main Bearing Fatigue Estimates From Turbulence Models

Main bearing fatigue damage was estimated according to Equation (13) for each 1-h simulation and averaged over the realizations of each condition. The bearing loads and low-speed shaft rotational speed, n_p , were extracted at each simulation step and the associated duration, t_p , equaled the simulation time step. The results are presented in Figure 12a. PyConTurb generally results in greater damage estimates, except in unstable, below-rated conditions, where Mann leads to more damage for

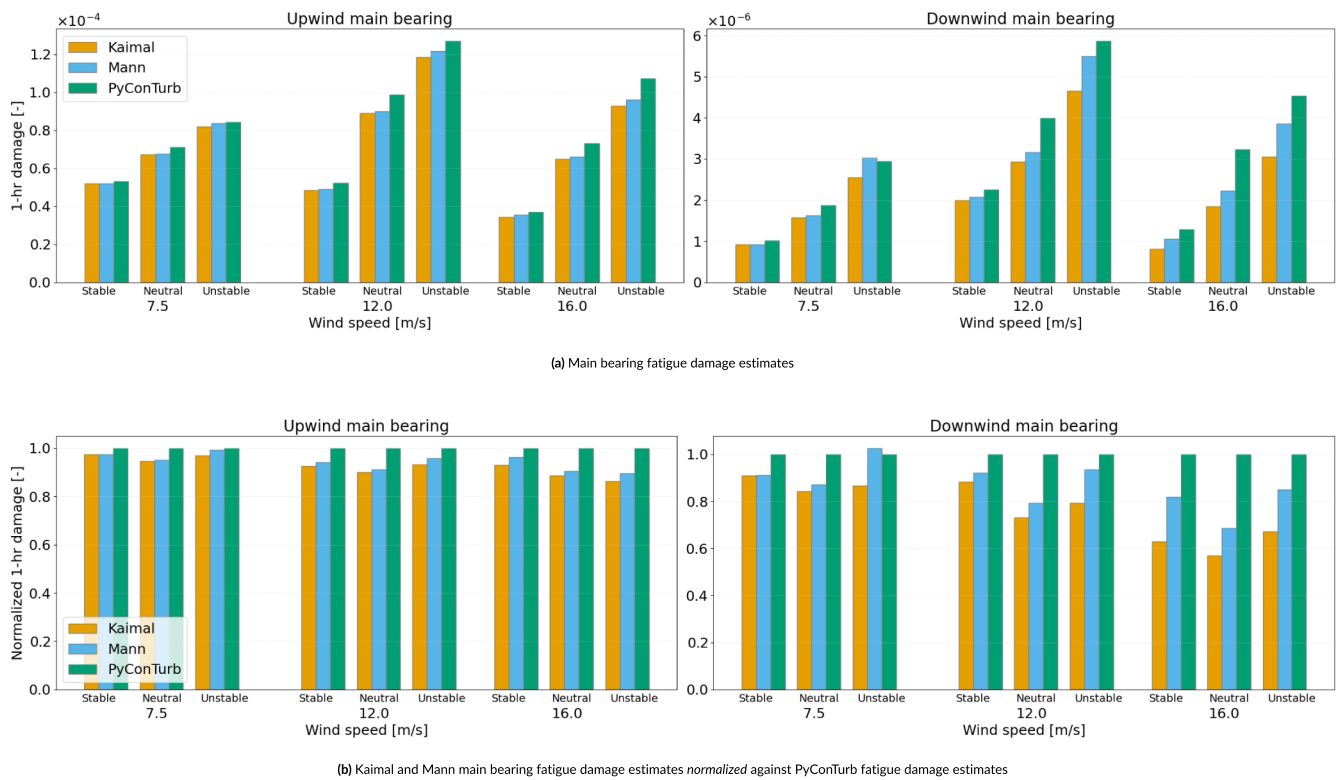


FIGURE 12 | Main bearing fatigue damage estimates.

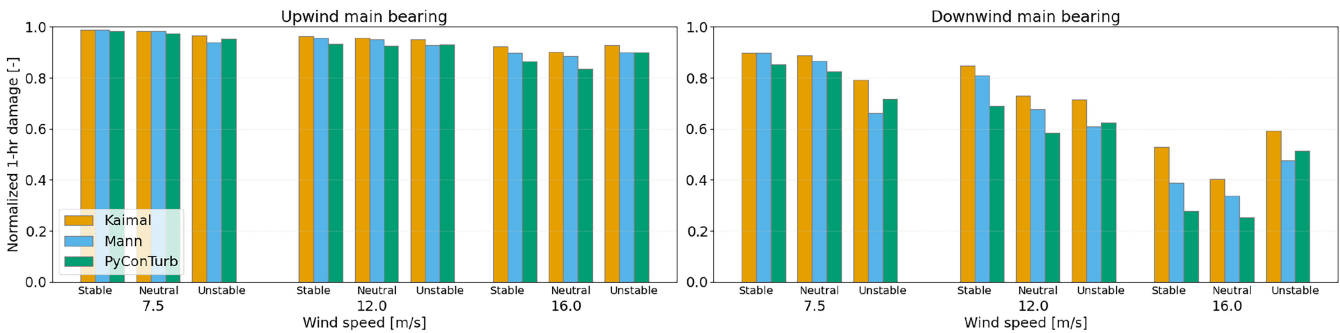


FIGURE 13 | Main bearing fatigue estimates from mean loads only normalized against total fatigue (the latter as shown in Figure 12a).

the downwind main bearing. Conversely, Kaimal gives the least conservative results, except in stable, below-rated conditions.

Looking at the damage normalized against PyConTurb results, Figure 12b, the relative difference between the models is largest for the downwind bearing, and increases with increasing wind speed. The biggest difference between the PyConTurb and Kaimal models is seen for the downwind main bearing in neutral conditions at 16-m/s mean wind speed, with Kaimal resulting in less than 60% of the PyConTurb fatigue estimates. At best, Kaimal can estimate 91% of the PyConTurb results for the upwind main bearing. Similarly, the Mann model estimates fatigue damage as low as 69% of PyConTurb results. For the upwind main bearing, Kaimal gives results on the order of 86% to 98% of PyConTurb. The significant differences among the atmospheric conditions can largely be attributed to variations in turbulence intensity and shear profile inherent to these canonical cases. Kenworthy et al. [17] pointed out the beneficial impact of

shear counteracting gravitational loads on main bearing rating life, which partly explains why stable conditions exhibit less fatigue damage than unstable. Kenworthy et al. further found that main bearing fatigue damage sensitivity to turbulence intensity was less significant. However, Figure 12 shows differences in damage when comparing Mann and Kaimal, both of which have the same shear profile, indicating that shear is not the only source of variations in main bearing damage.

4.2.2 | Influence of Time-Averaged Loads

For the bearing basic rating life, both the load fluctuations and the load magnitude matter (in contrast to rainflow-counting procedures applied for, e.g., the tower, where stress amplitudes are of main importance). This section aims to investigate the importance of static main bearing loads (a result of mean wind speed, wind shear and gravity) relative to dynamic loads (from

structural dynamics, turbulence, and waves) when considering main bearing fatigue.

Static bearing loads were obtained by time-averaging the loads over the entire time series. Fatigue damage from these time-averaged loads was then calculated according to Equation (13). In theory, X and Y should be calculated based on the time-averaged loads. However, to obtain comparability with the results from Section 4.2.1, the time series of X , Y , and n_i from Section 4.2.1 were applied here. The estimated fatigue was then normalized against the fatigue estimates from Section 4.2.1 and presented in Figure 13. This representation gives an indication of the relative importance of the mean and fluctuating wind velocity components for given conditions and turbulence models.

As shown in Figure 13, upwind bearing fatigue damage is dominated by mean loads. However, looking at the downwind main bearing, there are large variations among conditions in the contribution of mean loads, and at 16 m/s, dynamic loads seem as important as mean loads. Kaimal fatigue damage appears to be more governed by mean loads than damage estimated using Mann and PyConTurb. Based on these results, it is clear that differences in fatigue estimates between the turbulence models

cannot solely be described by differences in mean loads but also by the differences in turbulence formulations.

4.3 | Structural Response

Differences between turbulence models mainly influence the rotor aerodynamic moments governing the main bearing radial loads, whereas the axial loads are governed by rotor-averaged wind velocity. Thus, when investigating reasons for turbulence model impact on bearing fatigue in this section, focus is on radial loads.

4.3.1 | Radial Load Duration Distribution

Bearing radial loads are plotted against accumulated number of shaft revolutions (load duration distribution) to evaluate contributions to bearing damage, as described by Nejad et al. [61]. The number of revolutions was calculated using Equation (14). The radial loads of the 16-m/s mean wind speeds are shown, because these conditions exhibit the largest relative difference between the models (see Figure 14).

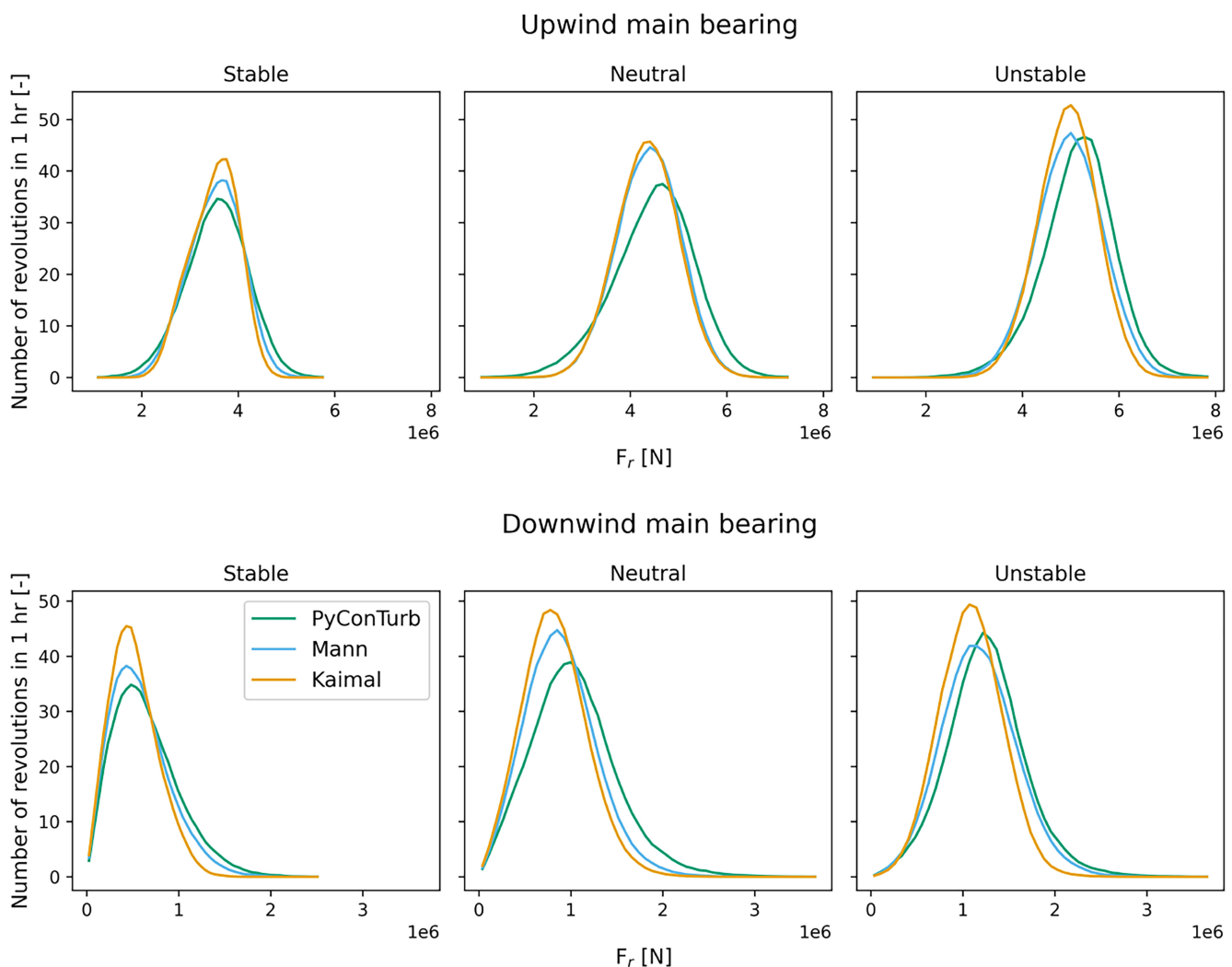


FIGURE 14 | Radial load duration distribution of the main bearings at 16-m/s conditions (EC3). Bin widths, y-axis and x-axis are shared along rows. Averaged results from all seeds are plotted.

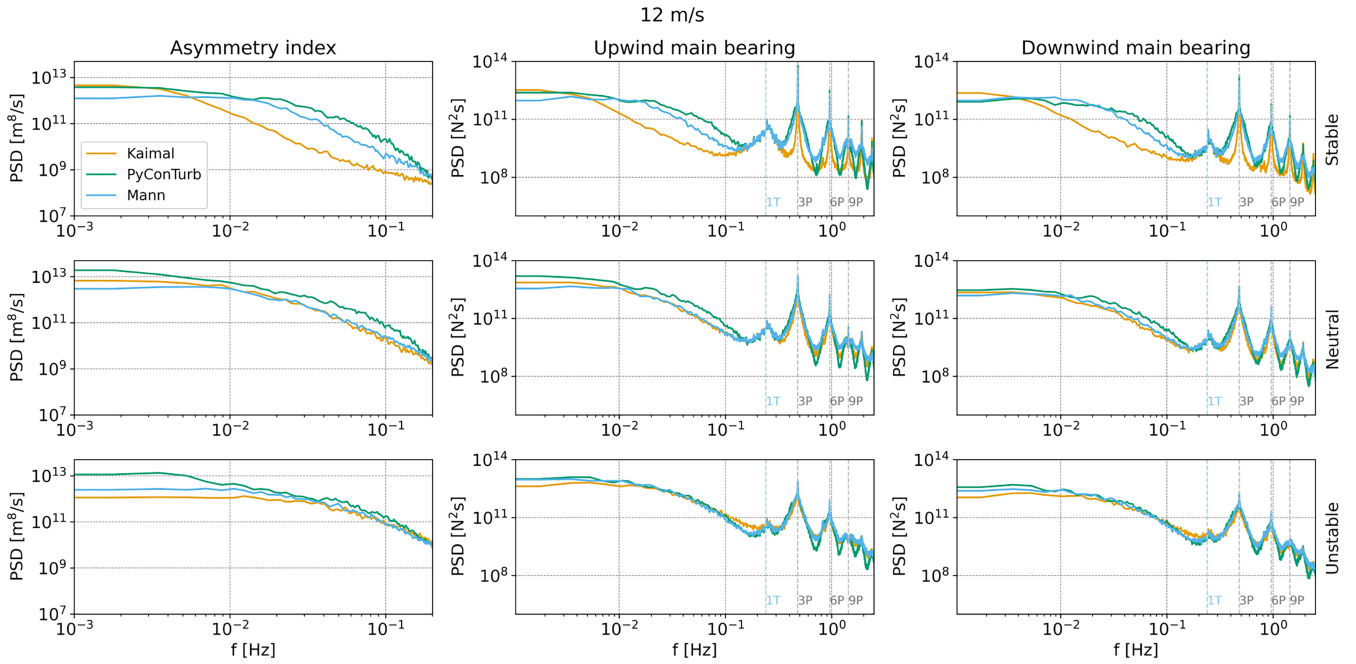


FIGURE 15 | Asymmetry index (left) and main bearing radial load response spectra (middle: upwind, right: downwind) averaged over seeds for each turbulence model and atmospheric condition at 12 m/s (EC2).

The upwind main bearing radial load distributions are more symmetric and exhibit larger values than the downwind main bearing. The downwind bearing histograms are more skewed, but the skewness decreases with instability. Bearing fatigue calculations effectively cube the loads, so that large loads cause more damage per revolution. The shift towards the right for the upwind main bearing indicates that this bearing experiences larger cumulative damage than the downwind main bearing. PyConTurb shows a wider spread in radial loads, whereas the Kaimal model results in the most narrow load distributions. PyConTurb exhibits larger radial loads than Kaimal and Mann, indicating that the bearings are subjected to a high radial load for more shaft revolutions. Generally, the peaks shift to higher radial loads when going from stable to neutral to unstable conditions.

4.3.2 | Bearing Load Spectra

Figure 15 presents bearing radial load spectra together with the spectra of the asymmetry index at 12 m/s. Similar figures for mean wind at 7.5 and 16 m/s are shown in Appendix C. Generally, the load response spectra have high energy in the low-frequency region, consistent with the turbulence spectrum. Then, an energy decay is seen before a peak at the first tower frequency (0.24 Hz). At higher frequencies, the multiples of the blade passing frequencies prevail.

The lower high-frequency energy of the PyConTurb turbulence spectra also appears in the high-frequency part of the bearing response spectra. However, at blade passing frequencies, PyConTurb shows as much or even more energy compared to Mann and Kaimal. Bearing response at multiples of the blade passing frequency stem from spatial coherence in the wind, mean shear and tower shadow. Hence, bearing loads at high frequencies can be caused by low-frequency variation in the wind.

For this reason, comparing frequencies in wind spectra and load spectra is complex. However, comparing bearing response spectra with the power spectra of the asymmetry index reveals clear similarities in the low-frequency region. Specifically, the spectra of the asymmetry index of PyConTurb also exhibit more energy in the low-frequency region than Mann and Kaimal.

In general, the load spectra of PyConTurb have more energy in the high-frequency peaks and in the low-frequency range, with Mann following and then Kaimal. This is consistent with the observed variations in main bearing fatigue estimates between the models. Moreover, the blade passing frequencies have relatively higher peaks for stable conditions than for neutral and unstable conditions compared to the low-frequency parts of the spectra.

5 | Conclusions

A coupled, medium-fidelity drivetrain model was implemented in OpenFAST, allowing for bending degrees of freedom and main bearing load output. The model was verified against a fully coupled multibody drivetrain-wind turbine model. With the new extended global model, accurate main bearing loads can be obtained directly from global analyses with low computational cost. The new model accommodates main bearing stiffness matrices with bending stiffness and off-diagonal terms. The accuracy of simple analytical equations compared to main bearing loads from such a coupled global model will be evaluated in future work.

Applying the new model, main bearing fatigue sensitivity to different turbulence models was investigated. Mann and Kaimal synthetic turbulence models were fitted to LES data for three atmospheric conditions (stable, neutral, and unstable) and three mean wind speeds (below, near, and above rated). Additionally,

PyConTurb, a constrained turbulence generator, was utilized to increase spatial resolution of the LES data. Main bearing rolling contact fatigue damage was computed for the different conditions and synthetic turbulence models. Using these methods, the following were found:

- Main bearing fatigue damage increased from stable to neutral to unstable atmospheric conditions.
- PyConTurb (LES) gave higher fatigue damage in most conditions.
- The difference in damage between the turbulence models was higher at higher wind speeds.
- Kaimal consistently underestimated bearing fatigue, with up to 40% less fatigue than PyConTurb.
- Mann also underestimated bearing fatigue (up to 30% less fatigue when compared to PyConTurb) but generally performed better than Kaimal.
- The upwind main bearing fatigue was highly governed by mean loads.
- The relative differences between turbulence models were most significant for the downwind main bearing.

Reasons for the differences between the models were investigated, with a focus on variations in the radial loads:

- Bearing fatigue damage is sensitive to mean loads from mean shear profile, but discrepancies in fatigue damage were seen between the Mann and Kaimal models, both of which exhibit the same shear profile. This indicates that dynamic loads are also important, especially for the downwind main bearing.
- Bearing radial load distributions were presented, showing that PyConTurb had a larger spread with higher loads, and Kaimal had the narrowest distributions.
- Main bearing radial load spectra showed that main bearing loads were highly driven by low-frequency content in the wind and turbulence sampling at blade passing frequencies. Bearing loads from PyConTurb had overall more energy.
- Relatively low co-coherence was expected to give increased bending moments on the shaft and therefore increased bearing loads. However, it was not trivial to directly link co-coherence to variations in bearing fatigue.
- An asymmetry index was calculated for the various turbulence fields, representing the large-scale asymmetry of the longitudinal wind velocity component over the rotor plane. The power spectra of the asymmetry index showed overall more variance for the PyConTurb wind fields than for Kaimal and Mann. The trends of the asymmetry index spectra closely resemble those of the low-frequency content of the bearing response spectra, suggesting that there is a correlation between bearing loads and asymmetric wind loads on the rotor at these frequencies.

To summarize, main bearing fatigue damage estimates were underestimated by synthetic turbulence models, likely because these models were not able to properly represent spatial variations

in the longitudinal component of the wind. However, one should read the results of this paper considering the assumptions made and limitations of the models used. Recommendations for future work include considering larger, floating turbines, the influence of drivetrain component flexibility on global and local response, the influence of wakes on main bearings, and considering more advanced methods for calculating bearing rating life.

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Peer Review

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A

Turbulence Spectra

Figure A1 shows turbulence spectra of the longitudinal wind velocity at hub height.

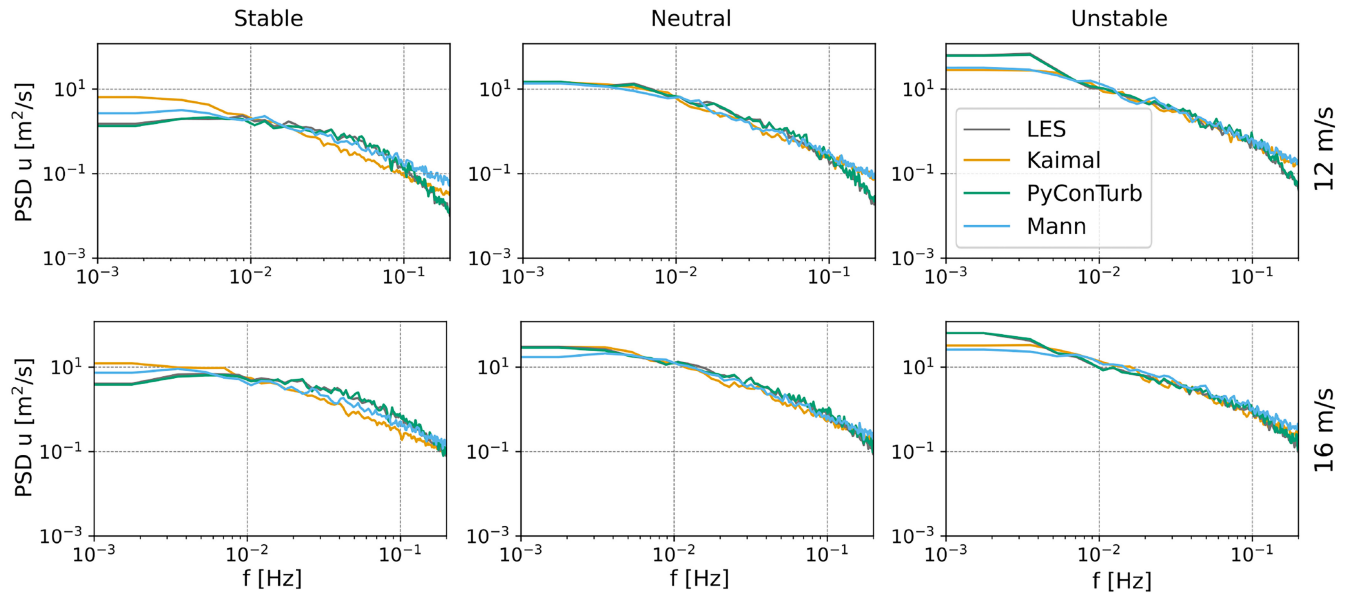


FIGURE A1 | Turbulence spectra of the longitudinal wind velocity component at hub height averaged over seeds for each turbulence model and atmospheric condition at 12 and 16 m/s.

Appendix B

Turbulence Co-Coherence

Figure B1 shows numerical co-coherence plotted against frequency for a lateral and vertical separation distance of about 0.7D, centered at the rotor.

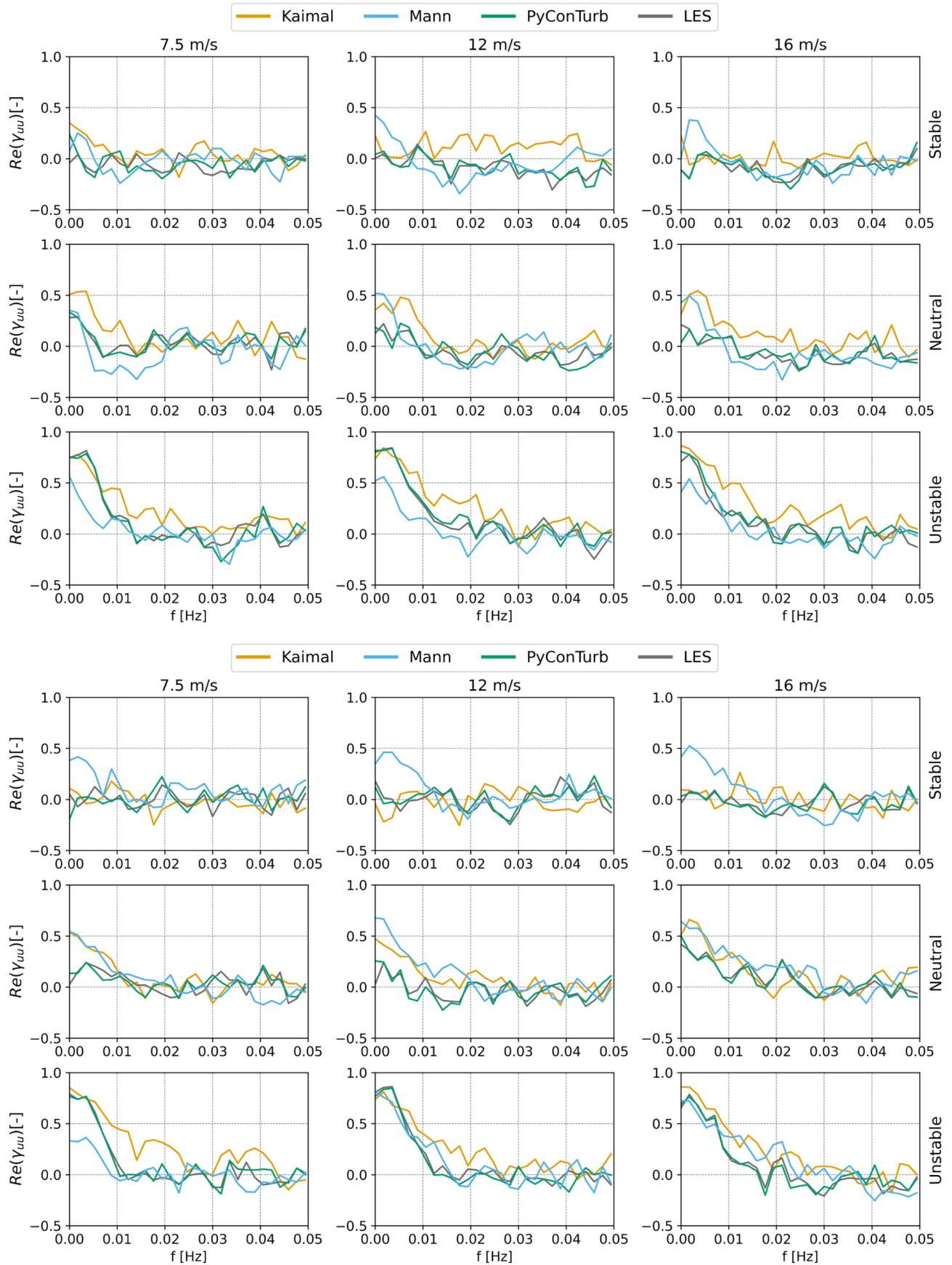


FIGURE B1 | Numerical co-coherence of the u -component for two points separated laterally (top) and vertically (bottom), centered at the hub height.

Appendix C

Spectra of Asymmetry Index and Main Bearing Radial Loads

Figures C1 and C2 present power spectral densities of the asymmetry index of the incoming wind field and the main bearing radial loads.

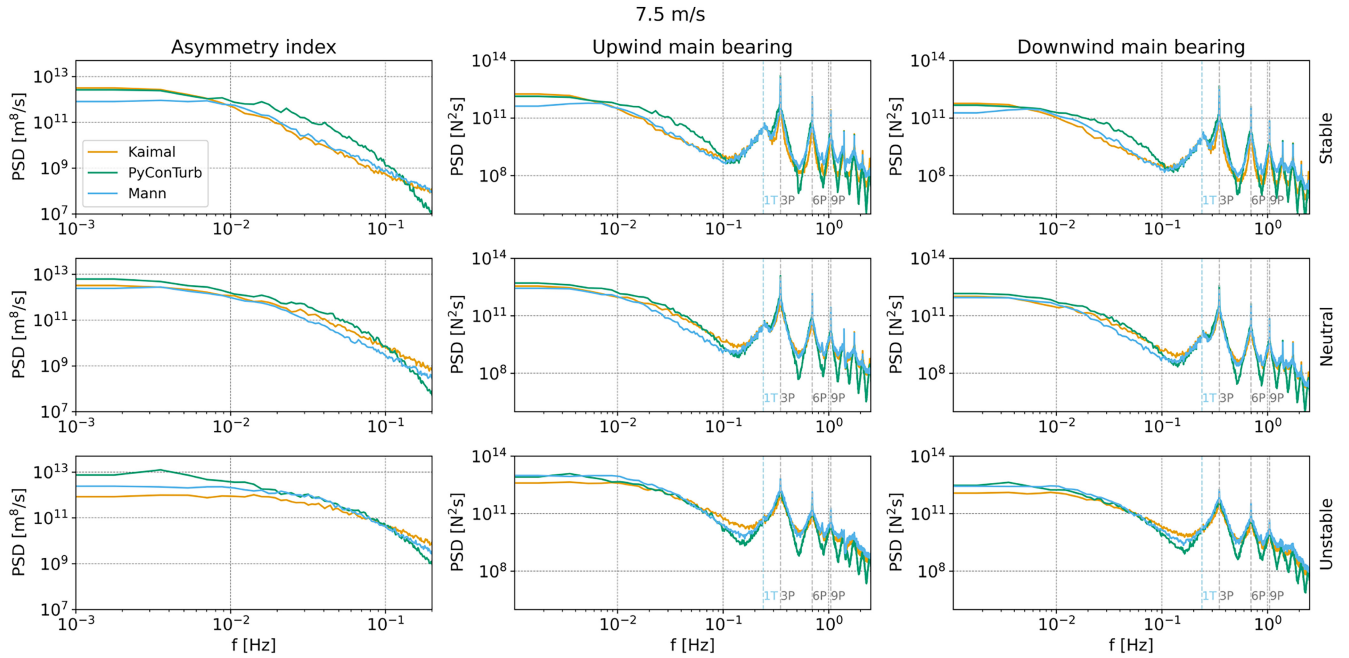


FIGURE C1 | Asymmetry index and main bearing radial load response spectra averaged over seeds for each turbulence model and atmospheric condition at 7.5 m/s.

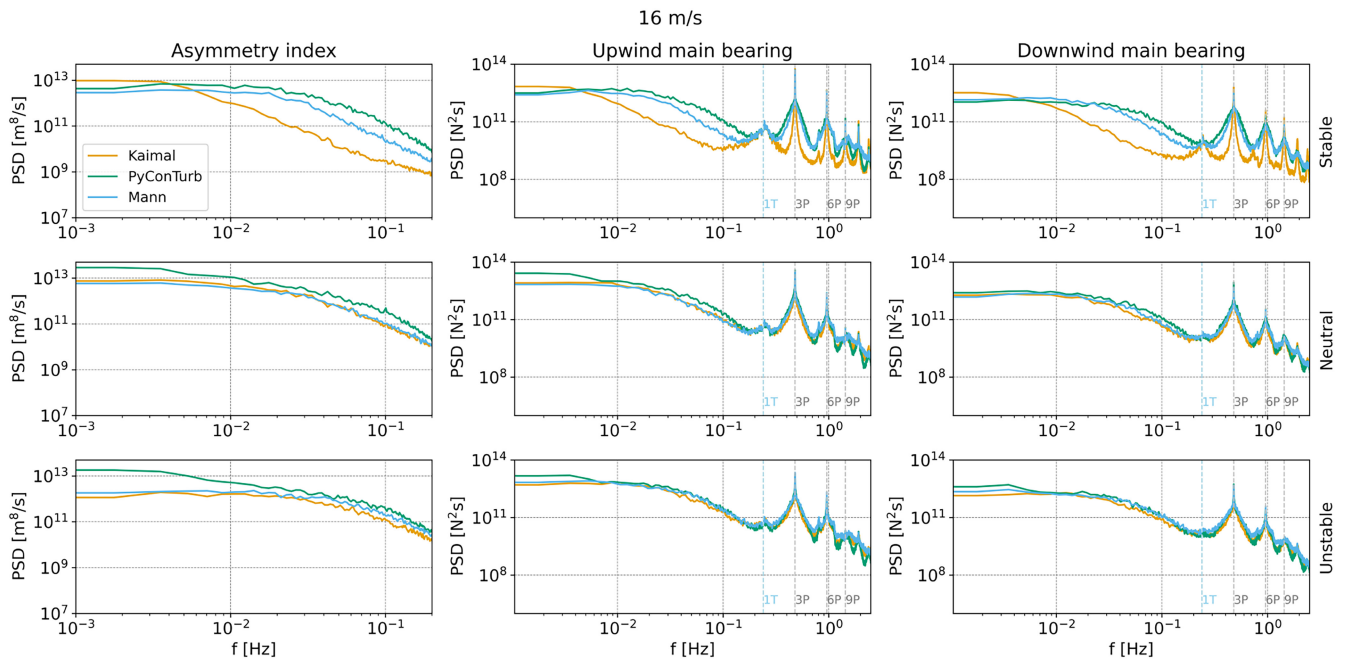


FIGURE C2 | Asymmetry index and main bearing radial load response spectra averaged over seeds for each turbulence model and atmospheric condition at 16 m/s.